THARUN YETTI

Software Engineer · Medical Coding & CDI Platforms · Healthcare RCM · Compliance-Driven Systems
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PROFESSIONAL SUMMARY

Software engineer with 1+ year building autonomous medical coding and Clinical Documentation Improvement (CDI) platforms - including AI-driven ICD-10/AR-DRG coding pipelines, real-time physician documentation-gap detection, and pre-submission audit workflows. Experienced in Python/FastAPI, AWS Bedrock/LLM integration, and HIPAA-aligned system design for regulated healthcare revenue cycle systems. Seeking to bring hands-on medical coding automation and compliance-first engineering to healthcare RCM technology

EXPERIENCE

Member of Technical Staff - *AmbitOS (AI Prof Private Ltd)* May 2026 - Present

- Built an autonomous AI medical coding pipeline using AWS Bedrock LLMs to extract diagnoses and procedure codes (ICD-10-AM, SBS/ACHI) with mandatory evidence citations back to source transcripts, eliminating manual chart extraction.
- Engineered a real-time Clinical Documentation Improvement (CDI) engine that analyzed live ambient consultation audio via WebSockets/Redis, flagging documentation gaps to physicians in-session and cutting retrospective coding queries.
- Developed a secondary clinical safety validation engine (hard stops, gender/age/code conflict checks) with automated re-validation on human coder edits, reducing downstream claim denials.
- Implemented an automated AR-DRG grouper computing Major Diagnostic Categories, relative cost weights, and reimbursement estimates from diagnoses, procedures, and length of stay.
- Built a coder/auditor workbench with real-time WebSocket queues, weighted search across 50,000+ ICD/SBS codes, and role-based multi-tenant governance with full audit trails
- Owned SIP/PSTN telephony integration (LiveKit, Twilio) and full AWS observability stack (Grafana, Prometheus, OpenTelemetry) supporting the platform's real-time patient-facing voice AI, maintaining <3% error rate at 500+ calls.
- Built and maintained CI/CD pipelines (GitHub Actions) with automated security scanning gating deployments from staging to production.

Full Stack Developer Intern - *Hitloop (now AmbitOS)* May 2025 - Apr 2026

- Contributed to a healthcare compliance platform covering PHI access control, HIPAA-aligned audit logging, and data residency - directly relevant to regulated healthcare data environments
- Built role-based dashboard features and integrated real-time voice AI into a patient-facing conversational agent.
- Built secure, scalable backend APIs in FastAPI, including cryptographic integrations.
- Cut real-time interaction latency ~30% by redesigning event handling and API flow.

SKILLS

Languages: Java, C, Python, JavaScript

Compliance & Security: HIPAA, GDPR, SOC2, ISO 27001, RBAC, Encryption (At-Rest/In-Transit), Audit Logging

Healthcare & Domain Standards: Medical Coding, ICD-10-AM, SBS (Saudi Billing System), ACHI, AR-DRG, Clinical Documentation Improvement (CDI), Condition Onset Flag (COF), Revenue Cycle Management (RCM)

Backend & Frameworks: FastAPI, Node.js, REST APIs, Spring Boot (Microservices)

AI/ML & Real-Time Systems: LLM Integration, Prompt Engineering, Scikit-learn, Pandas, NumPy

Voice AI & Telephony: LiveKit, WebRTC, SIP/Twilio, STT/TTS (Cartesia, Deepgram, OpenAI Realtime)

Frontend: React.js, Tailwind CSS, Material UI

DevOps & Cloud: Docker, Kubernetes, CI/CD, Microservices Architecture, Nginx, AWS

Databases & Caching: MySQL, PostgreSQL, MongoDB, Redis

Observability: Grafana, Prometheus, Loki, Promtail, OpenTelemetry, Langfuse, Alerting, CloudWatch

Tools & Platforms: Git, GitHub, IntelliJ, VSCode, Jupyter Notebook

EDUCATION

B.Tech in Computer Science, CGPA: 9.43 - *RGUKT, IIIT-RK Valley, AP, India* 2022 - 2026

Pre-University Course (Equivalent to Intermediate), CGPA: 9.94 - *RGUKT, IIIT-RK Valley, AP, India* 2020 - 2022

ACHIEVEMENTS

- 1st and 3rd place in campus hackathons (2025 & 2024).
- AIR 6009 in GATE CS 2025 with 519 score.
- 450+ DSA problems solved on LeetCode.